

# Package ‘fitVARMxID’

February 3, 2026

**Title** Fit the Vector Autoregressive Model for Multiple Individuals

**Version** 1.0.0

**Description** Fit the vector autoregressive model for multiple individuals using the 'OpenMx' package (Hunter, 2017 <[doi:10.1080/10705511.2017.1369354](https://doi.org/10.1080/10705511.2017.1369354)>).

**URL** <https://github.com/jeksterslab/fitVARMxID>,  
<https://jeksterslab.github.io/fitVARMxID/>

**BugReports** <https://github.com/jeksterslab/fitVARMxID/issues>

**License** MIT + file LICENSE

**Encoding** UTF-8

**Roxygen** list(markdown = TRUE)

**VignetteBuilder** knitr

**Depends** R (>= 4.1.0), OpenMx (>= 2.22.10)

**Imports** stats, methods, Matrix

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**RoxygenNote** 7.3.3.9000

**NeedsCompilation** no

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## Contents

|                           |    |
|---------------------------|----|
| coef.varmxid . . . . .    | 2  |
| converged . . . . .       | 3  |
| FitVARMxID . . . . .      | 4  |
| LDL . . . . .             | 13 |
| print.varmxid . . . . .   | 14 |
| Softplus . . . . .        | 15 |
| summary.varmxid . . . . . | 16 |
| vcov.varmxid . . . . .    | 18 |

|              |           |
|--------------|-----------|
| <b>Index</b> | <b>20</b> |
|--------------|-----------|

coef.varmxid

*Parameter Estimates***Description**

Parameter Estimates

**Usage**

```
## S3 method for class 'varmxid'
coef(
  object,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  ...
)
```

**Arguments**

|           |                                                                                                                                         |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------|
| object    | Object of class varmxid.                                                                                                                |
| mu        | Logical. If mu = TRUE, include estimates of the mu vector, if available. If mu = FALSE, exclude estimates of the mu vector.             |
| alpha     | Logical. If alpha = TRUE, include estimates of the alpha vector, if available. If alpha = FALSE, exclude estimates of the alpha vector. |
| beta      | Logical. If beta = TRUE, include estimates of the beta matrix, if available. If beta = FALSE, exclude estimates of the beta matrix.     |
| nu        | Logical. If nu = TRUE, include estimates of the nu vector, if available. If nu = FALSE, exclude estimates of the nu vector.             |
| psi       | Logical. If psi = TRUE, include estimates of the psi matrix, if available. If psi = FALSE, exclude estimates of the psi matrix.         |
| theta     | Logical. If theta = TRUE, include estimates of the theta matrix, if available. If theta = FALSE, exclude estimates of the theta matrix. |
| converged | Logical. Only include converged cases.                                                                                                  |
| grad_tol  | Numeric scalar. Tolerance for the maximum absolute gradient if converged = TRUE.                                                        |

|                 |                                                                                                                              |
|-----------------|------------------------------------------------------------------------------------------------------------------------------|
| hess_tol        | Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if converged = TRUE. |
| vanishing_theta | Logical. Test for measurement error variance going to zero if converged = TRUE.                                              |
| theta_tol       | Numeric. Tolerance for vanishing theta test if converged and theta_tol are TRUE.                                             |
| ...             | additional arguments.                                                                                                        |

**Value**

Returns a list of vectors of parameter estimates.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|           |                                |
|-----------|--------------------------------|
| converged | <i>Check Model Convergence</i> |
|-----------|--------------------------------|

---

**Description**

Evaluate whether OpenMx fit has converged successfully.

**Usage**

```
converged(object, ...)

## S3 method for class 'varmxid'
converged(
  object,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  prop = FALSE,
  ...
)
```

**Arguments**

|          |                                                                                                          |
|----------|----------------------------------------------------------------------------------------------------------|
| object   | An object of class varmxid.                                                                              |
| ...      | Passed to and/or used by methods.                                                                        |
| grad_tol | Numeric scalar. Tolerance for the maximum absolute gradient.                                             |
| hess_tol | Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value. |

|                 |                                                                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| vanishing_theta | Logical. Test for measurement error variance going to zero.                                                                                                                             |
| theta_tol       | Numeric. Tolerance for vanishing theta test.                                                                                                                                            |
| prop            | Logical. If prop = FALSE, a named logical vector indicating, for each individual fit, whether the convergence criteria are met. If prop = TRUE, the proportion of cases that converged. |

### Details

Convergence is defined by three criteria:

1. Status code equals 0L.
2. The maximum absolute gradient is below grad\_tol.
3. The Hessian is positive definite with all eigenvalues greater than hess\_tol.
4. If vanishing\_theta = TRUE, the model additionally checks that the diagonal elements of the measurement error covariance matrix ( $\Theta$ ) are not vanishingly small, where “small” is defined by theta\_tol.

### Value

For the varmxid method: If prop = FALSE, a named logical vector indicating, for each individual fit, whether the convergence criteria are met. If prop = TRUE, the proportion of cases that converged.

### Author(s)

Ivan Jacob Agaloos Pesigan

---

FitVARMxID

*Fit the First-Order Discrete-Time Vector Autoregressive Model by ID*

---

### Description

The function fits the first-order discrete-time vector autoregressive model for each unit ID.

### Usage

```
FitVARMxID(
  data,
  observed,
  id,
  ct = FALSE,
  time = NULL,
  center = TRUE,
  mu_fixed = FALSE,
  mu_free = NULL,
  mu_values = NULL,
  mu_lbound = NULL,
```

```
mu_ubound = NULL,  
alpha_fixed = FALSE,  
alpha_free = NULL,  
alpha_values = NULL,  
alpha_lbound = NULL,  
alpha_ubound = NULL,  
beta_fixed = FALSE,  
beta_free = NULL,  
beta_values = NULL,  
beta_lbound = NULL,  
beta_ubound = NULL,  
psi_diag = FALSE,  
psi_d_free = NULL,  
psi_d_values = NULL,  
psi_d_lbound = NULL,  
psi_d_ubound = NULL,  
psi_l_free = NULL,  
psi_l_values = NULL,  
psi_l_lbound = NULL,  
psi_l_ubound = NULL,  
nu_fixed = TRUE,  
nu_free = NULL,  
nu_values = NULL,  
nu_lbound = NULL,  
nu_ubound = NULL,  
theta_diag = TRUE,  
theta_fixed = TRUE,  
theta_d_free = NULL,  
theta_d_values = NULL,  
theta_d_lbound = NULL,  
theta_d_ubound = NULL,  
theta_d_equal = FALSE,  
theta_l_free = NULL,  
theta_l_values = NULL,  
theta_l_lbound = NULL,  
theta_l_ubound = NULL,  
mu0_fixed = TRUE,  
mu0_func = FALSE,  
mu0_free = NULL,  
mu0_values = NULL,  
mu0_lbound = NULL,  
mu0_ubound = NULL,  
sigma0_fixed = TRUE,  
sigma0_func = FALSE,  
sigma0_diag = FALSE,  
sigma0_d_free = NULL,  
sigma0_d_values = NULL,  
sigma0_d_lbound = NULL,
```

```

sigma0_d_ubound = NULL,
sigma0_l_free = NULL,
sigma0_l_values = NULL,
sigma0_l_lbound = NULL,
sigma0_l_ubound = NULL,
robust = FALSE,
tries_explore = 100,
tries_local = 10,
max_attempts = 10,
grad_tol = 0.01,
hess_tol = 1e-08,
eps = 1e-06,
factor = 10,
overwrite = FALSE,
path = getwd(),
prefix = "FitVARMxID",
seed = NULL,
silent = FALSE,
ncores = NULL,
clean = TRUE
)

```

### Arguments

|           |                                                                                                                                                                                           |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data      | Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), and at least one column of observed values. |
| observed  | Character vector. A vector of character strings of the names of the observed variables in the data.                                                                                       |
| id        | Character string. A character string of the name of the ID variable in the data.                                                                                                          |
| ct        | Logical. If TRUE, fit a continuous-time vector autoregressive model. If FALSE, fit a discrete-time vector autoregressive model.                                                           |
| time      | Character string. A character string of the name of the TIME variable in the data. Used when ct = TRUE.                                                                                   |
| center    | Logical. If TRUE, use the mean-centered (mean-reverting) state equation. Note that when center = TRUE, alpha is implied and the set-point mu is estimated.                                |
| mu_fixed  | Logical. If TRUE, the dynamic model set-point vector mu is fixed. If FALSE, mu is estimated.                                                                                              |
| mu_free   | Logical vector indicating which elements of mu are freely estimated. If NULL, all elements are free. Ignored if mu_fixed = TRUE.                                                          |
| mu_values | Numeric vector of values for mu. If mu_fixed = TRUE, these are fixed values. If mu_fixed = FALSE, these are starting values. If NULL, defaults to a vector of zeros.                      |
| mu_lbound | Numeric vector of lower bounds for mu. If NULL, no lower bounds are set. Ignored if mu_fixed = TRUE.                                                                                      |

|              |                                                                                                                                                                               |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| mu_ubound    | Numeric vector of upper bounds for mu. If NULL, no upper bounds are set. Ignored if mu_fixed = TRUE.                                                                          |
| alpha_fixed  | Logical. If TRUE, the dynamic model intercept vector alpha is fixed. If FALSE, alpha is estimated.                                                                            |
| alpha_free   | Logical vector indicating which elements of alpha are freely estimated. If NULL, all elements are free. Ignored if alpha_fixed = TRUE.                                        |
| alpha_values | Numeric vector of values for alpha. If alpha_fixed = TRUE, these are fixed values. If alpha_fixed = FALSE, these are starting values. If NULL, defaults to a vector of zeros. |
| alpha_lbound | Numeric vector of lower bounds for alpha. If NULL, no lower bounds are set. Ignored if alpha_fixed = TRUE.                                                                    |
| alpha_ubound | Numeric vector of upper bounds for alpha. If NULL, no upper bounds are set. Ignored if alpha_fixed = TRUE.                                                                    |
| beta_fixed   | Logical. If TRUE, the dynamic model coefficient matrix beta is fixed. If FALSE, beta is estimated.                                                                            |
| beta_free    | Logical matrix indicating which elements of beta are freely estimated. If NULL, all elements are free. Ignored if beta_fixed = TRUE.                                          |
| beta_values  | Numeric matrix of values for beta. If beta_fixed = TRUE, these are fixed values. If beta_fixed = FALSE, these are starting values. If NULL, defaults to a zero matrix.        |
| beta_lbound  | Numeric matrix of lower bounds for beta. If NULL, defaults to -1.5. Ignored if beta_fixed = TRUE.                                                                             |
| beta_ubound  | Numeric matrix of upper bounds for beta. If NULL, defaults to +1.5. Ignored if beta_fixed = TRUE.                                                                             |
| psi_diag     | Logical. If TRUE, psi is diagonal. If FALSE, psi is symmetric.                                                                                                                |
| psi_d_free   | Logical vector indicating free/fixed status of the elements of psi_d. If NULL, all element of psi_d are free.                                                                 |
| psi_d_values | Numeric vector with starting values for psi_d. If NULL, defaults to a vector of ones.                                                                                         |
| psi_d_lbound | Numeric vector with lower bounds for psi_d. If NULL, no lower bounds are set.                                                                                                 |
| psi_d_ubound | Numeric vector with upper bounds for psi_d. If NULL, no upper bounds are set.                                                                                                 |
| psi_l_free   | Logical matrix indicating which strictly-lower-triangular elements of psi_l are free. Ignored if psi_diag = TRUE.                                                             |
| psi_l_values | Numeric matrix of starting values for the strictly-lower-triangular elements of psi_l. If NULL, defaults to a null matrix.                                                    |
| psi_l_lbound | Numeric matrix with lower bounds for psi_l. If NULL, no lower bounds are set.                                                                                                 |
| psi_l_ubound | Numeric matrix with upper bounds for psi_l. If NULL, no upper bounds are set.                                                                                                 |
| nu_fixed     | Logical. If TRUE, the measurement model intercept vector nu is fixed. If FALSE, nu is estimated.                                                                              |
| nu_free      | Logical vector indicating which elements of nu are freely estimated. If NULL, all elements are free. Ignored if nu_fixed = TRUE.                                              |

|                |                                                                                                                                                                                             |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| nu_values      | Numeric vector of values for nu. If nu_fixed = TRUE, these are fixed values. If nu_fixed = FALSE, these are starting values. If NULL, defaults to a vector of zeros.                        |
| nu_lbound      | Numeric vector of lower bounds for nu. If NULL, no lower bounds are set. Ignored if nu_fixed = TRUE.                                                                                        |
| nu_ubound      | Numeric vector of upper bounds for nu. If NULL, no upper bounds are set. Ignored if nu_fixed = TRUE.                                                                                        |
| theta_diag     | Logical. If TRUE, theta is diagonal. If FALSE, theta is symmetric.                                                                                                                          |
| theta_fixed    | Logical. If TRUE, the measurement error matrix theta is fixed to SoftPlus(theta_d_values). If FALSE, only diagonal elements are estimated (off-diagonals fixed to zero).                    |
| theta_d_free   | Logical vector indicating free/fixed status of the diagonal parameters theta_d. If NULL, all element of theta_d are free.                                                                   |
| theta_d_values | Numeric vector with starting values for theta_d. If theta_fixed = TRUE, these are fixed values. If theta_fixed = FALSE, these are starting values. If NULL, defaults to an identity matrix. |
| theta_d_lbound | Numeric vector with lower bounds for theta_d. If NULL, no lower bounds are set.                                                                                                             |
| theta_d_ubound | Numeric vector with upper bounds for theta_d. If NULL, no upper bounds are set.                                                                                                             |
| theta_d_equal  | Logical. When TRUE, all free diagonal elements of theta_d are constrained to be equal and estimated as a single shared parameter (theta_eq). Ignored if no diagonal elements are free.      |
| theta_l_free   | Logical matrix indicating which strictly-lower-triangular elements of theta_l are free. Ignored if theta_diag = TRUE.                                                                       |
| theta_l_values | Numeric matrix of starting values for the strictly-lower-triangular elements of theta_l. If NULL, defaults to a null matrix.                                                                |
| theta_l_lbound | Numeric matrix with lower bounds for theta_l. If NULL, no lower bounds are set.                                                                                                             |
| theta_l_ubound | Numeric matrix with upper bounds for theta_l. If NULL, no upper bounds are set.                                                                                                             |
| mu0_fixed      | Logical. If TRUE, the initial mean vector mu0 is fixed. If FALSE, mu0 is estimated.                                                                                                         |
| mu0_func       | Logical. If TRUE and mu0_fixed = TRUE, mu0 is fixed to $(I - \beta)^{-1}\alpha$ .                                                                                                           |
| mu0_free       | Logical vector indicating which elements of mu0 are freely estimated.                                                                                                                       |
| mu0_values     | Numeric vector of values for mu0. If mu0_fixed = TRUE, these are fixed values. If mu0_fixed = FALSE, these are starting values. If NULL, defaults to a vector of zeros.                     |
| mu0_lbound     | Numeric vector of lower bounds for mu0. If NULL, no lower bounds are set. Ignored if mu0_fixed = TRUE.                                                                                      |
| mu0_ubound     | Numeric vector of upper bounds for mu0. If NULL, no upper bounds are set. Ignored if mu0_fixed = TRUE.                                                                                      |
| sigma0_fixed   | Logical. If TRUE, the initial covariance matrix sigma0 is fixed. If FALSE, sigma0 is estimated.                                                                                             |

|                              |                                                                                                                                                                                                                       |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>sigma0_func</code>     | Logical. If TRUE and <code>sigma0_fixed</code> = TRUE, <code>sigma0</code> is fixed to $(I - \beta \otimes \beta)^{-1} \text{Vec}(\Psi)$ .                                                                            |
| <code>sigma0_diag</code>     | Logical. If TRUE, <code>sigma0</code> is diagonal. If FALSE, <code>sigma0</code> is symmetric.                                                                                                                        |
| <code>sigma0_d_free</code>   | Logical vector indicating free/fixed status of the elements of <code>sigma0_d</code> . If NULL, all element of <code>sigma0_d</code> are free.                                                                        |
| <code>sigma0_d_values</code> | Numeric vector with starting values for <code>sigma0_d</code> . If NULL, defaults to a vector of ones.                                                                                                                |
| <code>sigma0_d_lbound</code> | Numeric vector with lower bounds for <code>sigma0_d</code> . If NULL, no lower bounds are set.                                                                                                                        |
| <code>sigma0_d_ubound</code> | Numeric vector with upper bounds for <code>sigma0_d</code> . If NULL, no upper bounds are set.                                                                                                                        |
| <code>sigma0_l_free</code>   | Logical matrix indicating which strictly-lower-triangular elements of <code>sigma0_l</code> are free. Ignored if <code>sigma0_diag</code> = TRUE.                                                                     |
| <code>sigma0_l_values</code> | Numeric matrix of starting values for the strictly-lower-triangular elements of <code>sigma0_l</code> . If NULL, defaults to a null matrix.                                                                           |
| <code>sigma0_l_lbound</code> | Numeric matrix with lower bounds for <code>sigma0_l</code> . If NULL, no lower bounds are set.                                                                                                                        |
| <code>sigma0_l_ubound</code> | Numeric matrix with upper bounds for <code>sigma0_l</code> . If NULL, no upper bounds are set.                                                                                                                        |
| <code>robust</code>          | Logical. If TRUE, calculate robust (sandwich) sampling variance-covariance matrix.                                                                                                                                    |
| <code>tries_explore</code>   | Integer. Number of extra tries for the wide exploration phase using <code>OpenMx::mxTryHardWideSearch()</code> with <code>checkHess</code> = FALSE.                                                                   |
| <code>tries_local</code>     | Integer. Number of extra tries for local polishing via <code>OpenMx::mxTryHard()</code> when gradients remain above tolerance.                                                                                        |
| <code>max_attempts</code>    | Integer. Maximum number of remediation attempts after the first Hessian computation fails the criteria. Each attempt may nudge off bounds, refit locally without the Hessian, and, on the last attempt, relax bounds. |
| <code>grad_tol</code>        | Numeric. Tolerance for the maximum absolute gradient. Smaller values are stricter.                                                                                                                                    |
| <code>hess_tol</code>        | Numeric. Minimum allowable Hessian eigenvalue. Smaller values are less strict.                                                                                                                                        |
| <code>eps</code>             | Numeric. Proximity threshold to detect parameters on their bounds and to nudge them inward by $10 * \text{eps}$ .                                                                                                     |
| <code>factor</code>          | Numeric. Multiplicative factor to relax parameter bounds on the final remediation attempt. Lower bounds are divided by <code>factor</code> and upper bounds are multiplied by <code>factor</code> .                   |
| <code>overwrite</code>       | Logical. If TRUE, existing intermediate files are overwritten. Defaults to FALSE.                                                                                                                                     |
| <code>path</code>            | Character string. Directory in which to save intermediate files.                                                                                                                                                      |

|        |                                                                              |
|--------|------------------------------------------------------------------------------|
| prefix | Alphanumeric character string. Prefix to use when naming intermediate files. |
| seed   | Random seed for reproducibility.                                             |
| silent | Logical. If TRUE, suppresses messages during the model fitting stage.        |
| ncores | Positive integer. Number of cores to use.                                    |
| clean  | Logical. If TRUE, clean intermediate files saved in path.                    |

## Details

The measurement model is given by

$$\mathbf{y}_{i,t} = \boldsymbol{\nu} + \boldsymbol{\Lambda}\boldsymbol{\eta}_{i,t} + \boldsymbol{\varepsilon}_{i,t}, \quad \text{with } \boldsymbol{\varepsilon}_{i,t} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Theta})$$

where  $\mathbf{y}_{i,t}$ ,  $\boldsymbol{\eta}_{i,t}$ , and  $\boldsymbol{\varepsilon}_{i,t}$  are random variables and  $\boldsymbol{\nu}$ ,  $\boldsymbol{\Lambda}$ , and  $\boldsymbol{\Theta}$  are model parameters.  $\mathbf{y}_{i,t}$  represents a vector of observed random variables,  $\boldsymbol{\eta}_{i,t}$  a vector of latent random variables, and  $\boldsymbol{\varepsilon}_{i,t}$  a vector of random measurement errors, at time  $t$  and individual  $i$ .  $\boldsymbol{\nu}$ , denotes a vector of intercepts (fixed to a null vector by default),  $\boldsymbol{\Lambda}$  a matrix of factor loadings, and  $\boldsymbol{\Theta}$  the covariance matrix of  $\boldsymbol{\varepsilon}$ . In this model,  $\boldsymbol{\Lambda}$  is an identity matrix and  $\boldsymbol{\Theta}$  is a diagonal matrix.

The dynamic structure is given by

$$\boldsymbol{\eta}_{i,t} = \boldsymbol{\alpha} + \boldsymbol{\beta}\boldsymbol{\eta}_{i,t-1} + \boldsymbol{\zeta}_{i,t}, \quad \text{with } \boldsymbol{\zeta}_{i,t} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Psi})$$

where  $\boldsymbol{\eta}_{i,t}$ ,  $\boldsymbol{\eta}_{i,t-1}$ , and  $\boldsymbol{\zeta}_{i,t}$  are random variables, and  $\boldsymbol{\alpha}$ ,  $\boldsymbol{\beta}$ , and  $\boldsymbol{\Psi}$  are model parameters. Here,  $\boldsymbol{\eta}_{i,t}$  is a vector of latent variables at time  $t$  and individual  $i$ ,  $\boldsymbol{\eta}_{i,t-1}$  represents a vector of latent variables at time  $t - 1$  and individual  $i$ , and  $\boldsymbol{\zeta}_{i,t}$  represents a vector of dynamic noise at time  $t$  and individual  $i$ .  $\boldsymbol{\alpha}$  denotes a vector of intercepts,  $\boldsymbol{\beta}$  a matrix of autoregression and cross regression coefficients, and  $\boldsymbol{\Psi}$  the covariance matrix of  $\boldsymbol{\zeta}_{i,t}$ .

If center = TRUE, the dynamic structure is parameterized as follows

$$\boldsymbol{\eta}_{i,t} = \boldsymbol{\mu}_{\boldsymbol{\eta}} + \boldsymbol{\beta}(\boldsymbol{\eta}_{i,t-1} - \boldsymbol{\mu}_{\boldsymbol{\eta}}) + \boldsymbol{\zeta}_{i,t}$$

where  $\boldsymbol{\mu}_{\boldsymbol{\eta}}$  is equilibrium level of the latent state toward which the system is pulled over time.

The continuous-time parameterization, when ct = TRUE, for the dynamic structure is given by

$$d\boldsymbol{\eta}_{i,t} = (\boldsymbol{\alpha} + \boldsymbol{\beta}\boldsymbol{\eta}_{i,t-1}) dt + \boldsymbol{\Psi}^{\frac{1}{2}} d\mathbf{W}_{i,t}$$

note that  $d\mathbf{W}$  is a Wiener process or Brownian motion, which represents random fluctuations.

If center = TRUE, the dynamic structure is parameterized as follows

$$d\boldsymbol{\eta}_{i,t} = \boldsymbol{\beta}(\boldsymbol{\eta}_{i,t-1} - \boldsymbol{\mu}_{\boldsymbol{\eta}}) dt + \boldsymbol{\Psi}^{\frac{1}{2}} d\mathbf{W}_{i,t}$$

## Value

Returns an object of class varmxid which is a list with the following elements:

**call** Function call.

**args** List of function arguments.

**fun** Function used ("FitVARMxID").

**output** A list of fitted OpenMx models.

**robust** A list of output from `OpenMx::mxRobustSE()` with argument details = TRUE for each id.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

Hunter, M. D. (2017). State space modeling in an open source, modular, structural equation modeling environment. *Structural Equation Modeling: A Multidisciplinary Journal*, 25(2), 307–324. doi:[10.1080/10705511.2017.1369354](https://doi.org/10.1080/10705511.2017.1369354)

Neale, M. C., Hunter, M. D., Pritikin, J. N., Zahery, M., Brick, T. R., Kirkpatrick, R. M., Estabrook, R., Bates, T. C., Maes, H. H., & Boker, S. M. (2015). OpenMx 2.0: Extended structural equation and statistical modeling. *Psychometrika*, 81(2), 535–549. doi:[10.1007/s1133601494358](https://doi.org/10.1007/s1133601494358)

**See Also**

Other VAR Functions: [LDL\(\)](#), [Softplus\(\)](#)

**Examples**

```
## Not run:
# Generate data using the simStateSpace package-----
set.seed(42)
k <- 2
n <- 5
time <- 100
alpha <- rep(x = 5, times = k)
beta <- matrix(
  data = c(.5, .0, .2, .5),
  nrow = k,
  ncol = k
)
psi <- matrix(
  data = c(exp(-4.1), exp(-3.9), exp(-3.9), exp(-3.2)),
  nrow = k,
  ncol = k
)
psi_l <- t(chol(psi))
nu <- rep(x = 0, times = k)
lambda <- diag(k)
theta <- matrix(
  data = c(exp(-2), 0, 0, exp(-2.8)),
  nrow = k,
  ncol = k
)
theta_l <- t(chol(theta))
mu0 <- c(solve(diag(k) - beta) %*% alpha)
sigma0 <- matrix(
  data = c(
    solve(diag(k * k) - beta %x% beta) %*% c(psi)
  ),
  nrow = k,
  ncol = k
)
```

```

)
sigma0_l <- t(chol(sigma0))
sim <- simStateSpace::SimSSMIVary(
  n = n,
  time,
  mu0 = list(mu0),
  sigma0_l = list(sigma0_l),
  alpha = list(alpha),
  beta = simStateSpace::SimBetaN(
    n = n,
    beta = beta,
    vcov_beta_vec_l = t(chol(0.1 * diag(k * k)))
  ),
  psi_l = list(psi_l),
  nu = list(nu),
  lambda = list(lambda),
  theta_l = list(theta_l)
)
data <- as.data.frame(sim)

# Fit the model-----
# center = TRUE
library(fitVARMxID)
fit <- FitVARMxID(
  data = data,
  observed = paste0("y", seq_len(k)),
  id = "id",
  center = TRUE
)
print(fit)
summary(fit)
coef(fit)
vcov(fit)
converged(fit)

# Fit the model-----
# center = FALSE
library(fitVARMxID)
fit <- FitVARMxID(
  data = data,
  observed = paste0("y", seq_len(k)),
  id = "id",
  center = FALSE
)
print(fit)
summary(fit)
coef(fit)
vcov(fit)
converged(fit)

## End(Not run)

```

---

LDL

*LDL' Decomposition of a Symmetric Positive-Definite Matrix*

---

## Description

Performs an LDL' factorization of a symmetric positive-definite matrix  $X$ , such that

$$X = LDL^{\top},$$

where  $L$  is unit lower-triangular (ones on the diagonal) and  $D$  is diagonal.

## Usage

`LDL(x)`

## Arguments

`x`                      Numeric matrix. Must be symmetric positive-definite.

## Details

This function returns both the unit lower-triangular factor  $L$  and the diagonal factor  $D$ . The strictly lower-triangular part of  $L$  is also provided for convenience. The function additionally computes an unconstrained vector `d_uc` such that `softplus(d_uc) = d_vec`, using `softplus-1(y) = log(exp(y) + 1)` for stable back-transformation.

## Value

A list with components:

|                           |                                                                 |
|---------------------------|-----------------------------------------------------------------|
| <code>l_mat_unit</code>   | Unit lower-triangular matrix $L$ .                              |
| <code>l_mat_strict</code> | Strictly lower-triangular part of $L$ .                         |
| <code>d_mat</code>        | Diagonal matrix $D$ .                                           |
| <code>d_vec</code>        | Vector of diagonal entries of $D$ .                             |
| <code>d_uc</code>         | Unconstrained vector with <code>softplus(d_uc) = d_vec</code> . |
| <code>x</code>            | Original input matrix.                                          |
| <code>y</code>            | Reconstructed matrix $LDL^{\top}$ .                             |
| <code>diff</code>         | Difference <code>x - y</code> .                                 |

## See Also

Other VAR Functions: [FitVARMxID\(\)](#), [Softplus\(\)](#)

## Examples

```
set.seed(123)
A <- matrix(rnorm(16), 4, 4)
S <- crossprod(A) + diag(1e-6, 4) # SPD
out <- LDL(S)
max(abs(out$diff))
```

---

print.varmxid

*Print Method for Object of Class varmxid*

---

## Description

Print Method for Object of Class varmxid

## Usage

```
## S3 method for class 'varmxid'
print(
  x,
  means = FALSE,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  digits = 4,
  ...
)
```

## Arguments

|       |                                                                                                                                         |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------|
| x     | an object of class varmxid.                                                                                                             |
| means | Logical. If means = TRUE, return means. Otherwise, the function returns raw estimates.                                                  |
| mu    | Logical. If mu = TRUE, include estimates of the mu vector, if available. If mu = FALSE, exclude estimates of the mu vector.             |
| alpha | Logical. If alpha = TRUE, include estimates of the alpha vector, if available. If alpha = FALSE, exclude estimates of the alpha vector. |

|                 |                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| beta            | Logical. If beta = TRUE, include estimates of the beta matrix, if available. If beta = FALSE, exclude estimates of the beta matrix.     |
| nu              | Logical. If nu = TRUE, include estimates of the nu vector, if available. If nu = FALSE, exclude estimates of the nu vector.             |
| psi             | Logical. If psi = TRUE, include estimates of the psi matrix, if available. If psi = FALSE, exclude estimates of the psi matrix.         |
| theta           | Logical. If theta = TRUE, include estimates of the theta matrix, if available. If theta = FALSE, exclude estimates of the theta matrix. |
| converged       | Logical. Only include converged cases.                                                                                                  |
| grad_tol        | Numeric scalar. Tolerance for the maximum absolute gradient if converged = TRUE.                                                        |
| hess_tol        | Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if converged = TRUE.            |
| vanishing_theta | Logical. Test for measurement error variance going to zero if converged = TRUE.                                                         |
| theta_tol       | Numeric. Tolerance for vanishing theta test if converged and theta_tol are TRUE.                                                        |
| digits          | Integer indicating the number of decimal places to display.                                                                             |
| ...             | further arguments.                                                                                                                      |

**Author(s)**

Ivan Jacob Agaloos Pesigan

Softplus

*Softplus and Inverse Softplus Transformations***Description**

The softplus transformation maps unconstrained real values to the positive real line. This is useful when parameters (e.g., variances) must be strictly positive. The inverse softplus transformation recovers the unconstrained value from a positive input.

**Usage**

Softplus(x)

InvSoftplus(x)

**Arguments**

x                      Numeric vector or matrix. Input values to be transformed.

**Details**

- $\text{Softplus}(x) = \log(1 + \exp(x))$
- $\text{InvSoftplus}(x) = \log(\exp(x) - 1)$

For numerical stability, these functions use `log1p()` and `expm1()` internally.

**Value**

- `Softplus()`: numeric vector or matrix of strictly positive values.
- `InvSoftplus()`: numeric vector or matrix of unconstrained values.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**See Also**

Other VAR Functions: [FitVARMxID\(\)](#), [LDL\(\)](#)

**Examples**

```
# Apply softplus to unconstrained values
x <- c(-5, 0, 5)
y <- Softplus(x)

# Recover unconstrained values
x_recovered <- InvSoftplus(y)

y
x_recovered
```

---

summary.varmxid

---

*Summary Method for Object of Class varmxid*


---

**Description**

Summary Method for Object of Class varmxid

**Usage**

```
## S3 method for class 'varmxid'
summary(
  object,
  means = FALSE,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
```

```

    nu = TRUE,
    psi = TRUE,
    theta = TRUE,
    converged = TRUE,
    grad_tol = 0.01,
    hess_tol = 1e-08,
    vanishing_theta = TRUE,
    theta_tol = 0.001,
    digits = 4,
    ...
)

```

### Arguments

|                 |                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| object          | an object of class varmxid.                                                                                                             |
| means           | Logical. If means = TRUE, return means. Otherwise, the function returns raw estimates.                                                  |
| mu              | Logical. If mu = TRUE, include estimates of the mu vector, if available. If mu = FALSE, exclude estimates of the mu vector.             |
| alpha           | Logical. If alpha = TRUE, include estimates of the alpha vector, if available. If alpha = FALSE, exclude estimates of the alpha vector. |
| beta            | Logical. If beta = TRUE, include estimates of the beta matrix, if available. If beta = FALSE, exclude estimates of the beta matrix.     |
| nu              | Logical. If nu = TRUE, include estimates of the nu vector, if available. If nu = FALSE, exclude estimates of the nu vector.             |
| psi             | Logical. If psi = TRUE, include estimates of the psi matrix, if available. If psi = FALSE, exclude estimates of the psi matrix.         |
| theta           | Logical. If theta = TRUE, include estimates of the theta matrix, if available. If theta = FALSE, exclude estimates of the theta matrix. |
| converged       | Logical. Only include converged cases.                                                                                                  |
| grad_tol        | Numeric scalar. Tolerance for the maximum absolute gradient if converged = TRUE.                                                        |
| hess_tol        | Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if converged = TRUE.            |
| vanishing_theta | Logical. Test for measurement error variance going to zero if converged = TRUE.                                                         |
| theta_tol       | Numeric. Tolerance for vanishing theta test if converged and theta_tol are TRUE.                                                        |
| digits          | Integer indicating the number of decimal places to display.                                                                             |
| ...             | further arguments.                                                                                                                      |

### Author(s)

Ivan Jacob Agaloos Pesigan

vcov.varmxid

*Sampling Covariance Matrix of the Parameter Estimates***Description**

Sampling Covariance Matrix of the Parameter Estimates

**Usage**

```
## S3 method for class 'varmxid'
vcov(
  object,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  robust = FALSE,
  ...
)
```

**Arguments**

|           |                                                                                                                                         |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------|
| object    | Object of class varmxid.                                                                                                                |
| mu        | Logical. If mu = TRUE, include estimates of the mu vector, if available. If mu = FALSE, exclude estimates of the mu vector.             |
| alpha     | Logical. If alpha = TRUE, include estimates of the alpha vector, if available. If alpha = FALSE, exclude estimates of the alpha vector. |
| beta      | Logical. If beta = TRUE, include estimates of the beta matrix, if available. If beta = FALSE, exclude estimates of the beta matrix.     |
| nu        | Logical. If nu = TRUE, include estimates of the nu vector, if available. If nu = FALSE, exclude estimates of the nu vector.             |
| psi       | Logical. If psi = TRUE, include estimates of the psi matrix, if available. If psi = FALSE, exclude estimates of the psi matrix.         |
| theta     | Logical. If theta = TRUE, include estimates of the theta matrix, if available. If theta = FALSE, exclude estimates of the theta matrix. |
| converged | Logical. Only include converged cases.                                                                                                  |
| grad_tol  | Numeric scalar. Tolerance for the maximum absolute gradient if converged = TRUE.                                                        |

|                 |                                                                                                                                               |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| hess_tol        | Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if converged = TRUE.                  |
| vanishing_theta | Logical. Test for measurement error variance going to zero if converged = TRUE.                                                               |
| theta_tol       | Numeric. Tolerance for vanishing theta test if converged and theta_tol are TRUE.                                                              |
| robust          | Logical. If TRUE, use robust (sandwich) sampling variance-covariance matrix. If FALSE, use normal theory sampling variance-covariance matrix. |
| ...             | additional arguments.                                                                                                                         |

**Value**

Returns a list of sampling variance-covariance matrices.

**Author(s)**

Ivan Jacob Agaloos Pesigan

# Index

## \* VAR Functions

FitVARMxID, [4](#)

LDL, [13](#)

Softplus, [15](#)

## \* fitVARMxID

FitVARMxID, [4](#)

LDL, [13](#)

Softplus, [15](#)

## \* fit

FitVARMxID, [4](#)

## \* methods

coef.varmxid, [2](#)

converged, [3](#)

print.varmxid, [14](#)

summary.varmxid, [16](#)

vcov.varmxid, [18](#)

## \* misc

LDL, [13](#)

Softplus, [15](#)

coef.varmxid, [2](#)

converged, [3](#)

FitVARMxID, [4](#), [13](#), [16](#)

InvSoftplus (Softplus), [15](#)

LDL, [11](#), [13](#), [16](#)

OpenMx::imxRobustSE(), [10](#)

print.varmxid, [14](#)

Softplus, [11](#), [13](#), [15](#)

summary.varmxid, [16](#)

vcov.varmxid, [18](#)